

A Guide to Statistical Network Modeling

Session 3.2: SAOMs for Selection and Influence

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instats Seminar

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Companion computations and reproducibility

- **Shared resources:** [repository](#) and [companion app](#); see Session 1.1 for the common reproducibility workflow.
- Open **Session 3.2** and choose one of **five visible public coevolution workflows**; each states its reuse disclosure, behavior-support rule, estimates, convergence diagnostics, simulation envelopes, and downloadable record.
- **Computation boundary:** the app fits a stated joint RSiena network–behavior SAOM; network selection and behavioral influence parameters are conditional model associations, not automatic causal effects.

Learning objectives

By the end of this session, we will be able to:

- define a **joint network–behavior state** and distinguish network selection from behavioral influence;
- specify alter, ego, and similarity effects in a network evaluation function;
- specify linear shape, quadratic shape, and average-similarity effects in a behavior evaluation function;
- state why coevolution estimates do not automatically resolve homophily, common shocks, or causal peer effects; and
- audit joint RSiena fits with behavior-distribution and network-structure simulation checks.

Session route

- ① **Coevolution objects:** repeated networks, repeated behavior, and two coupled microprocesses.
- ② **Selection model:** behavior as an actor, alter, and similarity covariate for tie choices.
- ③ **Influence model:** network exposure in the actor's behavior-choice process.
- ④ **Estimation and auditing:** joint moments, convergence, and all public-app plots.
- ⑤ **No-code practice:** five documented public behavior workflows plus constrained BYOD input.

A joint network–behavior state

- Let $X(t)$ be the repeated binary network and let $V(t) = (V_i(t))$ be a repeated actor behavior on a documented scale.
- A **joint SAOM** treats $(X(t), V(t))$ as the state of one stochastic process.
- At a network opportunity, an actor may alter one outgoing tie; at a behavior opportunity, an actor may move one permitted behavior step or remain unchanged.
- The processes are coupled because behavior can enter tie evaluation and network position can enter behavior evaluation.

Selection and influence are distinct model components

- **Selection** means that observed behavior values enter the *network* objective: behavior is associated with whom actors choose as alters.
- **Influence** means that network relations enter the *behavior* objective: an actor's behavior change is associated with its network neighborhood.
- The words describe locations in the model, not proven causal mechanisms.
- Both components may be present, absent, poorly identified, or confounded with omitted processes in the same empirical setting.

Behavior response and measurement scale

- Session 3.2 assumes a repeatedly measured, finite-valued actor behavior $V_i(t)$ with a common meaning at all selected waves.
- In RSiena terminology this is called a **behavioral dependent variable**, although substantively it may represent an actor-level state, attitude, score, or behavior.
- Examples include ordered substance-use or delinquency scores in public school-network panels.
- The admissible behavior choice set must respect the observed scale bounds; it is not an unrestricted continuous outcome.
- A score's numeric coding should be substantively defensible: changing the coding can change similarity and shape-effect interpretation.

Behavior-change opportunities

- Actor i receives behavior-change opportunities at rate $\lambda_i^V(x, v; \rho^V)$.
- A standard RSiena fit includes **period-specific behavior rate parameters** alongside network rate parameters.
- At a behavior opportunity, the allowed alternatives typically include the current score and adjacent permitted scores on the declared scale.
- A rate is not an observed count of questionnaire changes; it is a process parameter used to simulate the unobserved interval.

The behavior evaluation function

- The **behavior evaluation function** can be written as

$$f_i^V(x, v; \gamma) = \sum_{\ell=1}^L \gamma_{\ell} s_{i\ell}^V(x, v).$$

- It ranks actor i 's permitted behavior alternatives conditional on the current network and behavior state.
- Like the network objective, it is a local conditional-choice function, not a linear regression equation for observed behavior levels.
- The joint model links the two objectives through explicitly named cross-process statistics.

Network selection: behavior as an alter effect

- The **behavior alter effect** places potential alter j 's score V_j in actor i 's network evaluation function.
- Informally, it asks whether high- versus low-valued alters are relatively more attractive as tie targets after other included effects are held fixed.
- Its change statistic concerns the behavior of the *recipient* of a potential tie, not the ego's subsequent behavior change.
- A nonzero alter effect may reflect opportunity, status, measurement, shared contexts, or other omitted processes as well as a substantive preference.

Network selection: behavior as an ego effect

- The **behavior ego effect** places actor i 's own score V_i in its network evaluation function.
- It asks whether actors at different behavior levels tend to create more or fewer outgoing ties, conditional on the stated model.
- It is distinct from popularity: an ego effect concerns sent ties, whereas an alter effect concerns received tie choices.
- Behavior-based activity differences should not be called influence because the behavior enters the network process, not the behavior process.

Network selection: behavior similarity

- A common bounded similarity statistic is

$$\text{sim}(V_i, V_j) = 1 - \frac{|V_i - V_j|}{\Delta_V},$$

where Δ_V is the declared range used to normalize the behavior variable.

- RSiena's precise similarity construction follows the software definition; a sample's observed extrema need not define the substantive scale range.
- The **behavior similarity effect** rewards potential ties to actors with similar observed behavior values.
- It operationalizes homophily within the stated scale and support; it does not distinguish preference for similarity from a common cause of tie and behavior.
- Similarity is undefined for a constant behavior; the app rejects a selected behavior with no variation.

Selection effects act during tie choices

- For a possible tie $i \rightarrow j$, alter, ego, and similarity statistics can each change the local network objective.
- Their parameters are conditional on density, reciprocity, closure, and one another.
- The fitted selection component cannot be interpreted by examining a correlation between an adjacency matrix and a behavior vector alone.
- Structural network terms remain necessary because unmodeled reciprocity or closure can otherwise be attributed incorrectly to a behavior covariate.

Behavior linear shape

- The **linear shape effect** captures a baseline directional tendency in the behavior-choice process over its allowable scale.
- It is a within-model tendency toward higher or lower scores, conditional on the other behavior and network terms.
- The sign depends on coding orientation; reversing score labels reverses the interpretation.
- It is not the observed sample mean and should be interpreted together with the quadratic shape effect.

Behavior quadratic shape

- The **quadratic shape effect** allows the baseline behavior objective to curve across the permitted score range.
- It can represent a tendency away from, or toward, middle versus extreme values after accounting for the linear term.
- Its interpretation depends on centering and the discrete behavior scale implemented by the data and software.
- A poorly supported extreme category can make this term unstable; inspect the observed behavior distribution at every wave.

Behavioral influence: average similarity

- The **average-similarity effect** places network-neighbor behavior similarity in the behavior evaluation function.
- For actor i , it summarizes similarity between V_i and the behavior of the alters to whom i is tied, according to the explicitly specified directed-network convention.
- A positive parameter makes adjacent behavior moves that increase similarity to relevant network neighbors relatively more attractive in the stated process.
- It is a model-based influence parameter; it does not alone identify a causal peer effect.

Why similarity is not a causal proof

- Similar behavior among connected actors can arise before observed ties, through shared contexts, common exposures, selection on unmeasured traits, or genuine social influence.
- Repeated observations and joint modeling make these alternatives explicit, but do not automatically eliminate them.
- Causal interpretation requires assumptions about timing, confounding, measurement, interference, and the relevant population.
- The app therefore reports the effect as **average-similarity association in the specified behavior objective**.

Joint moment conditions

- Joint estimation compares observed and simulated statistics from both processes.
- Let $m_X(y, v)$ collect network statistics and $m_V(y, v)$ collect behavior statistics; the target is

$$\mathbb{E}_\theta \left\{ \begin{bmatrix} m_X(X, V) \\ m_V(X, V) \end{bmatrix} \right\} \approx \begin{bmatrix} m_X(x_{\text{obs}}, v_{\text{obs}}) \\ m_V(x_{\text{obs}}, v_{\text{obs}}) \end{bmatrix}.$$

- Network and behavior effects are estimated together because the simulated process changes both objects.
- More parameters require more information, not merely a longer coefficient table.

What a complete behavior support requires

- A coevolution calculation requires one observed behavior value for every retained actor at every selected wave.
- The app first applies the all-wave network support rule, then restricts public behavior workflows to actors with complete selected behavior observations.
- The app displays the number excluded for incomplete behavior; it never substitutes zero, carries a value forward, or creates a score.
- For BYOD input, incomplete behavior causes rejection rather than silent case filtering.

Behavior missingness is a modeling issue

- A missing behavior value may reflect nonresponse, out-of-range coding, survey design, or actor absence; these mechanisms are not interchangeable.
- Restricting to complete cases can change the target population and may induce selection bias.
- Imputation, composition handling, or a missing-data model require explicit assumptions and must be documented separately.
- The no-code app uses a conservative, transparent complete-behavior support rule rather than fabricating an unobserved behavior.

Network composition and behavior composition

- Network actor support and behavior completeness are separate conditions.
- An actor can be observed in the network at all selected waves but lack a behavior score at one wave.
- A behavior-complete coevolution panel is therefore not necessarily the same population as the network-only Session 3.1 panel.
- Report both restrictions whenever comparing network-only and coevolution results from the same public release.

Model specification in the companion app

- **Network objective:** density, directed reciprocity, directed transitive triplets, behavior alter, behavior ego, and behavior similarity.
- **Behavior objective:** linear shape, quadratic shape, and average network similarity.
- **Rates:** RSiena estimates period-specific network and behavior rate parameters.
- The specification is displayed before computation; the app does not offer a hidden automatic model search.
- The five worked workflows reuse this same declared baseline so that support and behavior differences are inspectable.

Convergence for joint SAOMs

- Joint fits add behavior effects and behavior rates to the convergence problem.
- **Apply the Session 3.1 convergence audit to every joint parameter**, including behavior rates, selection, shape, and influence effects: inspect all final t -ratios and the maximum ($|t| < 0.25$, preferably < 0.10).
- Instability can signal sparse behavior categories, weak variation, excessive complexity, incorrect support, or insufficient simulation settings.

Coefficient plot in the app

- The app displays each RSiena estimate with an approximate 95% Wald interval using estimate $\pm 1.96 \times$ standard error.
- These intervals are conditional on the fitted specification and the estimation approximation.
- A wide interval signals limited precision; it does not convert a noncausal association into a null causal effect.
- Interpret rate, network, selection, shape, and influence parameters in their distinct model locations.

Convergence-ratio plot in the app

- The app extends the Session 3.1 convergence display to every joint parameter; use the same ± 0.25 reference and investigate values near or beyond it rather than deleting effects selectively.
- This is an estimation audit, not a goodness-of-fit or causal-identification check.

Network goodness-of-fit plots in the app

- **Network GOF:** reuse the full Session 3.1 network-simulation audit—out-/in-degree, directed triads, reciprocity, geodesic distance, closure/two-path structure, and component/isolate summaries—as applicable, now using simulations from the fitted joint model.
- Read the network checks with the behavior and joint network–behavior diagnostics that follow. Simulation envelopes diagnose fitted-process variation; they are not a single accept/reject test or evidence of causal identification.

Behavior-distribution goodness-of-fit plot

- The app compares the observed behavior-score frequency distribution with distributions from simulated fitted states.
- It also compares the observed distribution of behavior changes with fitted simulations and displays behavior transitions across successive waves.
- A joint network–behavior audit compares tied-actor behavior mixing and similarity summaries with fitted simulations.
- These checks assess whether the behavior shape and average-similarity components reproduce observed behavior and its network coupling, not merely network moments.
- A model may fit network degree patterns while failing to represent behavior prevalence or extremes.
- Always inspect this plot together with the behavior mean-by-wave descriptive plot displayed before estimation.

All plots in the Session 3.2 app

- **Dynamics:** retain the Session 3.1 observed-dynamics, estimation, convergence, and support-appropriate network-simulation audit for the network component of the joint fit.
- **Selection:** observed ego-, alter-, and difference-specific tie rates; an ego $\times$ alter mixing heatmap; selection-effect intervals; a fitted selection surface; and observed-versus-simulated mixing, homophily, tied-actor similarity, and network-behavior association summaries.
- **Influence:** behavior-score distributions, mean and variance, transitions and individual changes; ego versus average-alter scores; change versus average-alter score and ego-alter discrepancy; influence/shape intervals and fitted contributions; and behavior distribution/change simulation checks.
- **Scope:** **Dynamics** asks whether network evolution is reproduced; **Selection** asks whether the model reproduces who chooses whom; **Influence** asks whether it reproduces behavior change relative to network exposure.
- No plot alone establishes causal influence: observed plots are descriptive, interval/convergence plots audit estimation, and envelopes are simulation-based goodness-of-fit diagnostics.

Worked public workflow 1: Knecht friendship–delinquency

- **Data:** public four-wave Dutch classroom friendship network with repeated delinquency score.
- **Support:** actors observed at all selected waves with complete selected delinquency values.
- **Question encoded by the model:** how are tie choices conditionally associated with delinquency ego, alter, and similarity, and how is delinquency change conditionally associated with network-neighbor similarity?
- **Disclosure:** this is a model-based coevolution association in one classroom panel, not a causal peer-delinquency estimate.

Worked public workflow 2: Knecht friendship–alcohol

- **Data:** the same documented classroom network release, now with alcohol scores available from waves 2–4.
- **Support:** the app uses only waves 2–4 because alcohol is unavailable at wave 1 by source design.
- **Why retain a second workflow?** it demonstrates that a distinct repeated behavior changes the valid observation range and behavior distribution.
- **Reuse disclosure:** this is not an independent study; it is a different transparent coevolution specification on the same public panel.

Worked public workflow 3: Glasgow friendship–alcohol

- **Data:** full public Glasgow friendship network and repeated alcohol score.
- **Support:** the app begins with the documented all-wave network selection, then excludes actors lacking alcohol at any selected wave and displays that count.
- **Response:** binary friendship presence retains source friend codes 1 and 2; structural absence code 10 is never treated as zero.
- **Limit:** complete-case restriction changes the modeled cohort and must be reported with the fitted result.

Worked public workflow 4: Glasgow friendship–cannabis

- **Data:** the same full public Glasgow network release with repeated cannabis score.
- **Support:** the behavior-completeness restriction is applied separately because missingness can differ from the alcohol workflow.
- **Use:** compare observed score distributions, selected actor support, and the stated joint model without pretending that cannabis and alcohol are interchangeable behaviors.
- **Reuse disclosure:** this is a second behavior specification from the same public release, not a second independent population.

Worked public workflow 5: s50 friendship–smoking

- **Data:** the public 50-actor, three-wave RSiena teaching excerpt with repeated smoking score.
- **Support:** all supplied actors and waves are retained; the numerical panel is compact enough for a reproducible classroom run.
- **Use:** inspect every coefficient, convergence, and simulation plot on a small documented example before uploading a new panel.
- **Limit:** the source is a teaching excerpt rather than a fully delineated population study.

No-code workflow for a public coevolution panel

- 1 Open the Render-hosted **Session 3.2** page and select a visible workflow.
- 2 Read the source, support rule, reuse disclosure, response definition, and interpretation limit.
- 3 Inspect dynamics, selection, and influence descriptive panels, including behavior distributions and support exclusions.
- 4 Choose seed, n_3 , and goodness-of-fit simulations, then run the stated RSiena computation.
- 5 Audit component-specific coefficients, all convergence ratios, network/selection/influence envelopes, and the joint network–behavior association checks before interpretation.

Bring Your Own Data: required CSV structure

- **Nodes CSV:** `wave,id`; **edges CSV:** `wave,source,target`; both must define a repeated loopless binary network.
- **Behavior CSV:** `wave,id` plus at least one numeric repeated behavior column; choose the column explicitly in the app.
- The upload is rejected if behavior values are missing after the balanced network panel is formed.
- Declare network directionality honestly; Session 3.2 selection/influence baseline is designed for directed friendship-like relations.
- Retain the downloadable record with source description, coding, support decisions, settings, estimates, and diagnostics.

Minimum reporting record for coevolution SAOMs

- State network population, support/composition rules, tie response, behavior coding, wave dates, and all missing-data handling.
- Separate network selection effects from behavior influence effects in the specification table.
- Report simulation settings, seed, convergence diagnostics, warnings, and whether estimates are classroom pilots or final analysis.
- Show dynamics, selection, influence, and joint network–behavior simulation diagnostics, including mismatches and complete-case exclusions.
- State explicitly which causal assumptions are *not* supplied by the fitted association model.

Session 3.2 synthesis

- Coevolution SAOMs represent repeated networks and repeated actor behavior as coupled actor-oriented microprocesses.
- Selection effects reside in the network objective; influence effects reside in the behavior objective.
- Similarity can arise from several mechanisms, so a parameter label does not establish causal peer influence.
- Complete behavior support, joint convergence assessment, and network–behavior simulation diagnostics are necessary parts of the calculation.
- Transparent data boundaries and limitations are stronger scientific practice than automated causal claims.

References



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